

ECO102: Introduction to Macroeconomics

Midterm | Spring 2026

Duration: 1 hr 15 mins | Total marks: 30

Section A- Numerical Problems (15 Marks)

Answer All the questions in this section. Show all your workings. When applicable, give your answers in two-decimal places.

1. The fictional economy of "Novaria" produces wheat flour, bread, tractors and coffee beans. Assuming 2024 is the base year, the table below summarizes the quantities and market prices for 2024 and 2025:

Economic Activity	2024 Quantity	2024 Price (\$)	2025 Quantity	2025 Price (\$)
Wheat flours used for baking bread	1,000	10	1,200	12
Households purchased breads for consumption	500	40	600	50
Local businesses invested in buying tractors	10	5,000	12	5,500
Domestically produced coffee Beans were exported	200	15	220	20

Given the information above, calculate the following.

- Calculate the Nominal GDP and the Real GDP for the year 2025, using the expenditure approach. [3 marks]
 - Calculate the percentage change in Nominal GDP between year 2024 and 2025. [3 marks]
 - Calculate the Real GDP Growth Rate between 2024 and 2025. [2 marks]
2. In the country of Econopolis, assume that households only consume four goods: Rice, Cheese, Laptops and Coffee. The following table presents the household purchases of these goods for 2022, 2023, and 2024. Assume 2022 as the base year

Goods Consumed	2022 (Base Year)		2023		2024	
	Price (\$)	Quantity	Price (\$)	Quantity	Price (\$)	Quantity
Rice	10	150	12	135	15	130
Cheese	15	50	20	40	25	30
Laptops	100	10	90	15	80	20
Coffee	20	15	30	10	26	12

- Calculate the Consumer Price Index (CPI) for the years 2023 and 2024. Show all necessary workings. [4 marks]
- Calculate the inflation rate for both 2023 and 2024 and interpret your results. Show the formula used and all workings. [3 marks]

Section B – Analytical Problems (15 Marks)

Answer any two questions in this section. When applicable, draw properly labeled diagrams to support your answer.

- The government of Maryland wants to boost the economy after the pandemic crisis. They have decided to reduce the income tax on wage earners from 10% to 8%. How does a decrease in income tax affect the labor market and potential GDP? Briefly discuss and depict using appropriate diagrams. [7.5 marks]
- According to the Bureau of Statistics of Country X, the working-age population (15–64 years) has increased by 4.88% over the past ten years, reaching 65.5% (10.82 crore) of the total population. Recently, the country has experienced a strong economic recovery, leading to increased job creation and a significant fall in the unemployment rate. Discuss how this situation might affect the real GDP and potential GDP of Country X. [7.5 marks]
- In Country X, which is heavily dependent on oil as a production input, the ongoing geopolitical conflict has caused an oil shortage. Since oil is essential for machinery, transport, and energy-intensive production processes, workers cannot work as efficiently as before without adequate fuel and power. How does this impact the labor market model and the production function? Briefly discuss and illustrate using appropriate diagrams. [7.5 marks]